IBM Data Science Capstone Project

SpaceX Falcon 9 Launch Analysis
Predicting First-Stage Landing Success

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IBM Data Science Professional Certificate | October 2026

Tools: Python, SQL, Pandas, Scikit-learn, Folium, Plotly and Dash

https://github.com/RAHULKR123456/IBM-SpaceX-Capstone

Executive Summary

Methods

- Data: SpaceX API request failed (HTTP 525), so IBM course datasets were used; Wikipedia revision scraped
- Prepared 90 launches: **60 landed**, **30 did not**
- EDA (Matplotlib, Seaborn), SQL (SQLite), Folium map, Plotly and Dash views
- Four classifiers tuned with 5-fold grid search; 80/20 stratified split (72 train, 18 test)

Key results

- Landing success: **0%** (2010 to 2013) to **90.0%** (2019) and **84.2%** (2020)
- Best site: KSC LC-39A **76.9%**; lowest: CCAFS LC-40 **26.9%**
- Payload band 2,000 to 3,999 kg has the highest success rate (61.9%); no single payload threshold
- Best model: **SVM, 83.3% test accuracy** vs 66.7% baseline; finds every success, half of the failures

Objective: investigate factors associated with first-stage landing success and assess whether historical launch data can predict the outcome.

Introduction

Project background

- SpaceX reuses the Falcon 9 first stage to reduce the cost of spaceflight
- After separation, the booster can land on a ground pad or a drone ship and be recovered
- A successful landing directly affects the price of a launch

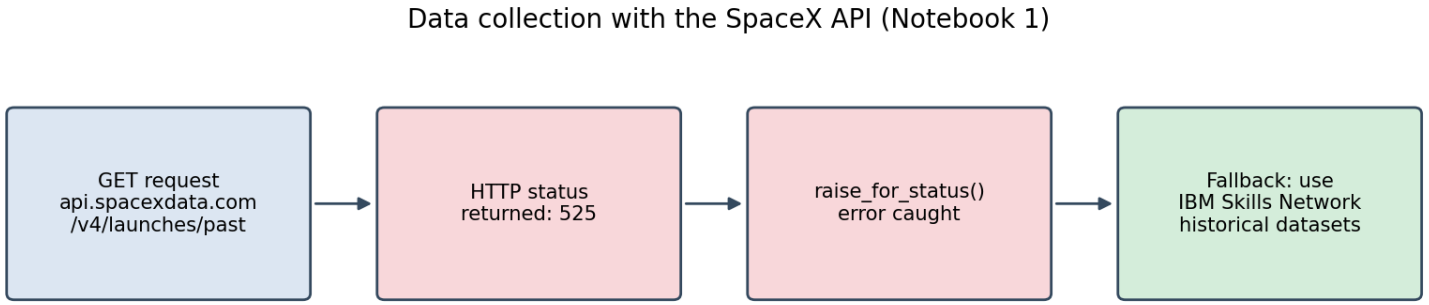
Problem statement

Can historical launch information (site, payload mass, orbit, booster configuration, flight history) help estimate the probability that the first stage will land successfully?

Research questions

- RQ1: How have landing outcomes changed over time and across launch sites?
- RQ2: Is landing success related to payload mass or orbit?
- RQ3: Which classification model predicts landing success best?

Data Collection: SpaceX API



API call flow (Notebook 1)

Endpoint: https://api.spacexdata.com/v4/launches/past

GitHub: https://github.com/RAHULKR123456/IBM-SpaceX-Capstone/blob/main/01_Data_Collection_API.ipynb

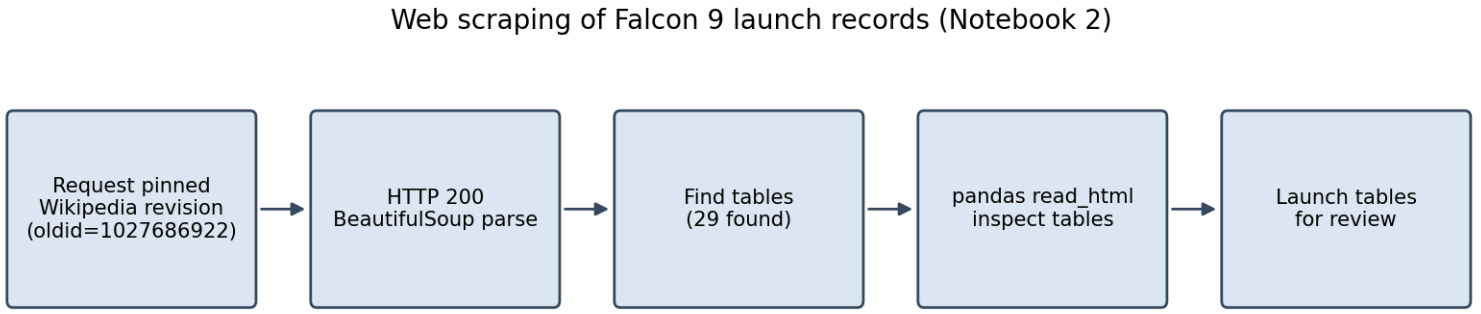
Key phrases

- requests.get(url, timeout=30)
- response.raise_for_status()
- pd.json_normalize(launches_json)

Outcome

HTTP status **525**; error caught and printed. Historical IBM course datasets used as the fallback.

Data Collection: Web Scraping



Web scraping flow (Notebook 2)

Key phrases

- Pinned revision: oldid=1027686922 (9 June 2021)
- HTTP 200 with a User-Agent header
- BeautifulSoup parse: **29 tables** found
- pandas.read_html; table shapes inspected

Source: Wikipedia, List of Falcon 9 and Falcon Heavy launches. A fixed revision keeps the results reproducible.

GitHub: https://github.com/RAHULKR123456/IBM-SpaceX-Capstone/blob/main/02_Data_Collection_Wikipedia.ipynb

Data Wrangling Methodology

Step	Method
Inspect	Shape, head, data types, missing-value counts
Clean	Strip column names; convert PayloadMass to numeric; fill missing values (mean in Notebook 3, median in Notebook 7)
Target Class	1 = first stage landed, 0 = did not; in Notebook 7 the label is built from the original outcomes and checked against the feature rows
Encode	One-hot encode Orbit, LaunchSite, LandingPad and Serial: 83 numeric features
Save	Export dataset_part_2_cleaned.csv for later notebooks

Target distribution: **60 landed (Class 1)** and **30 did not (Class 0)**, a 2:1 class imbalance that matters when evaluating models.

GitHub: https://github.com/RAHULKR123456/IBM-SpaceX-Capstone/blob/main/03_Data_Wrangling.ipynb

EDA with Data Visualization: Methodology

Chart	Purpose
Bar chart: launches by launch site	Compare how launches are split across sites
Categorical scatter: flight number vs launch site	See when each site was used and how outcomes changed
Scatter: payload mass vs flight number	Look for a relationship between payload and landing result
Bar chart: success rate by orbit	Rank orbit types by landing success
Line chart: success rate by year	Show the trend over time

Tools: Matplotlib, Seaborn and Pandas on the 90-launch table (Notebook 4). Points are coloured by landing outcome (0 = failure, 1 = success).

GitHub: https://github.com/RAHULKR123456/IBM-SpaceX-Capstone/blob/main/04_EDA_Visualization.ipynb

EDA with SQL: Methodology

- **Notebook 5:** SPACEXTBL (101 rows, 10 columns) loaded into SQLite; queries for unique launch sites, mission outcomes by site and minimum payload mass
- **Notebook 5b:** dashboard dataset (56 launches) queried for launches and success rate per site, site ranking, payload statistics, payload bands, overall totals and yearly success rate
- Aggregation (COUNT, SUM, AVG, MIN, MAX), GROUP BY, ORDER BY and RANK() window function
- SQL summaries were compared with the Python analysis

Tool: SQLite with pandas.read_sql_query

GitHub: https://github.com/RAHULKR123456/IBM-SpaceX-Capstone/blob/main/05_EDA_SQL.ipynb | https://github.com/RAHULKR123456/IBM-SpaceX-Capstone/blob/main/05b_EDA_SQL_Additional_Queries.ipynb

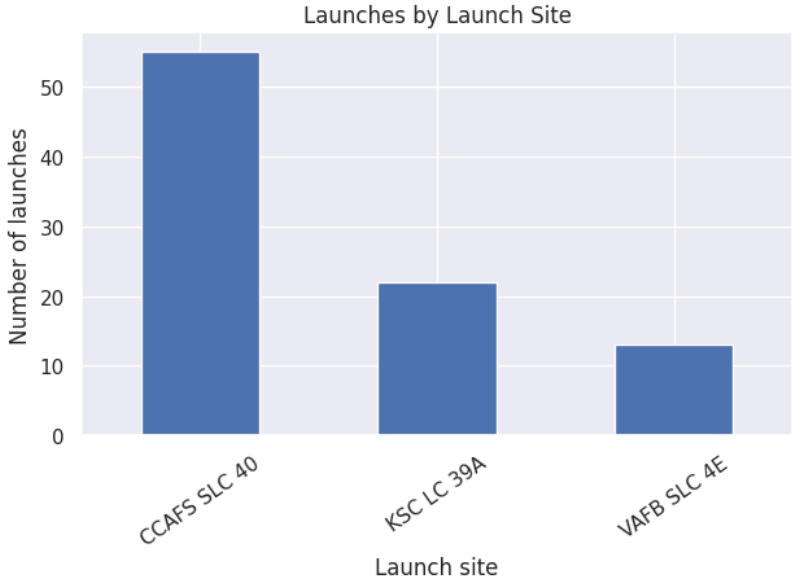
Interactive Visual Analytics: Methodology

Tool	What was built	File
Folium	Interactive map of the 4 launch sites with 56 launch records in a marker cluster (green = success, red = failure)	06_Folium_Interactive_Map.ipynb
Plotly Express	Success-rate bar chart by site and payload vs outcome scatter plot	08_Interactive_Dashboard_Plots.ipynb
Dash	Dashboard with site dropdown, success pie chart, payload range slider and payload vs outcome scatter	app.py

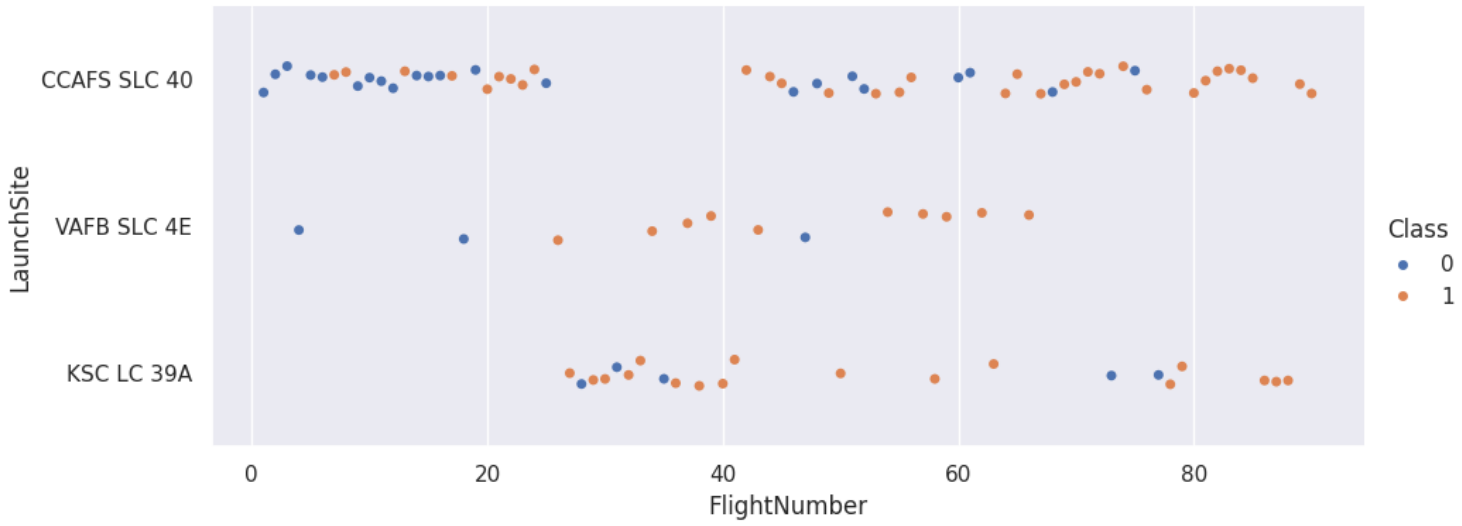
Purpose: let users explore the launch data geographically and compare sites and payloads interactively.

GitHub: https://github.com/RAHULKR123456/IBM-SpaceX-Capstone/blob/main/06_Folium_Interactive_Map.ipynb | https://github.com/RAHULKR123456/IBM-SpaceX-Capstone/blob/main/08_Interactive_Dashboard_Plots.ipynb | https://github.com/RAHULKR123456/IBM-SpaceX-Capstone/blob/main/app.py

EDA Results: Launch Sites and Flight Number



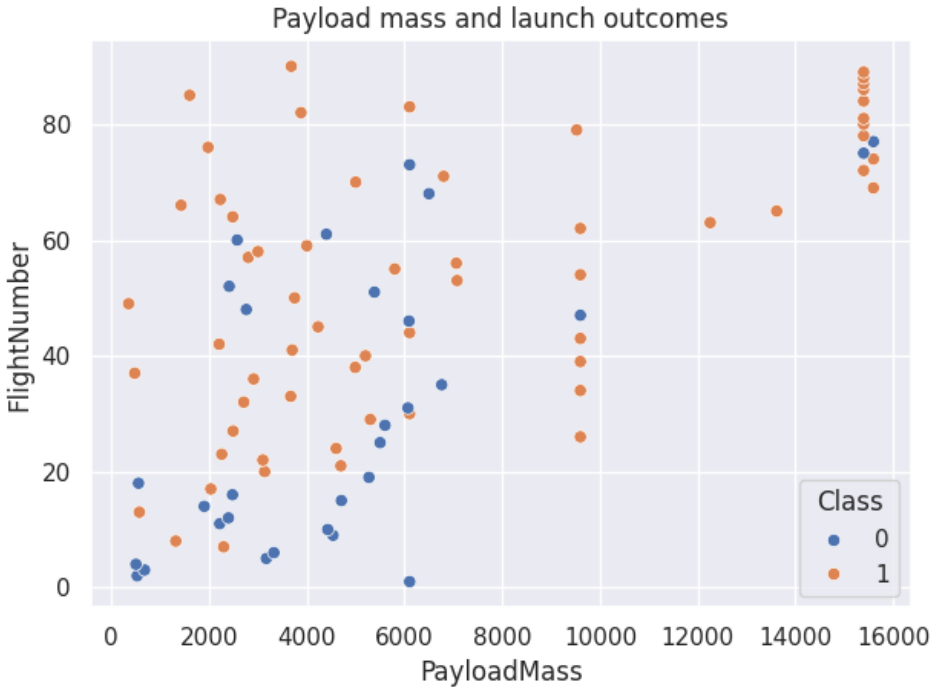
Launches by launch site (bar chart)



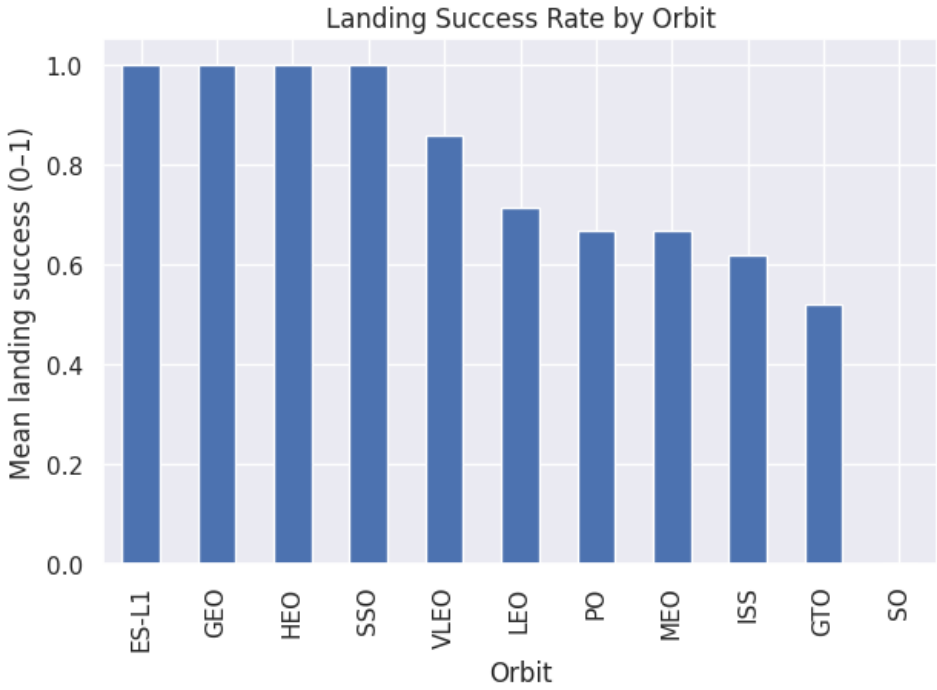
Flight number vs launch site, coloured by outcome (0 = failure, 1 = success)

- CCAFS SLC 40 hosted the most launches (about 55), then KSC LC 39A (about 22) and VAFB SLC 4E (about 13)
- At CCAFS SLC 40 the early flights are mostly failures and later flights mostly successes
- KSC LC 39A is used only from about flight 26 onward and is mostly successful

EDA Results: Payload Mass and Orbit



Payload mass vs flight number, coloured by outcome



Landing success rate by orbit

- Heavy payloads (about 15,500 kg) occur late and are mostly successful; most failures occur at lower flight numbers with payloads below about 7,000 kg
- No single payload threshold separates successes from failures
- ES-L1, GEO, HEO and SSO reach 100% on few launches; GTO is lowest among common orbits (51.9%); SO is 0%

EDA Results: Yearly Trend



Landing success rate by year

Year	Success rate
2010	0.000
2012	0.000
2013	0.000
2014	0.333
2015	0.333
2016	0.625
2017	0.833
2018	0.611
2019	0.900
2020	0.842

- Success was 0% up to 2013
- Rose to 83.3% in 2017, dipped to 61.1% in 2018
- Reached 90.0% in 2019 and 84.2% in 2020
- The clearest pattern in the data: success is tied to how far the programme had progressed

EDA with SQL Results (1): Launch Sites, Outcomes, Payload

Query 1: unique launch sites

```
SELECT DISTINCT "Launch_Site" FROM SPACEXTBL
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Query 3: minimum payload

```
SELECT MIN("PAYLOAD_MASS__KG_") FROM SPACEXTBL
```

MinimumPayloadMass
0 kg

Query 2: mission outcomes by launch site

```
SELECT "Launch_Site","Mission_Outcome",COUNT(*) FROM SPACEXTBL
GROUP BY 1,2
```

Launch site	Mission outcome	Count
CCAFS LC-40	Failure (in flight)	1
CCAFS LC-40	Success	25
CCAFS SLC-40	Success	32
CCAFS SLC-40	Success (separate row)	1
CCAFS SLC-40	Success (payload unclear)	1
KSC LC-39A	Success	25
VAFB SLC-4E	Success	16

Meaning

- Four site names: three in Florida, one in California
- Almost every mission was a success; the only in-flight failure is at CCAFS LC-40
- Lightest payload: 0 kg (early test flights)

GitHub: https://github.com/RAHULKR123456/IBM-SpaceX-Capstone/blob/main/05_EDA_SQL.ipynb

EDA with SQL Results (2): Success Rates, Rankings, Payloads

Queries 4 and 5: success rate and ranking by site

```
SELECT RANK() OVER (ORDER BY 1.0*SUM(Class)/COUNT(*) DESC) AS Rank,
LaunchSite, ... FROM LAUNCHES GROUP BY LaunchSite
```

Rank	LaunchSite	Launches	Successes	SuccessRate Pct
1	KSC LC-39A	13	10	76.9%
2	CCAFS SLC-40	7	3	42.9%
3	VAFB SLC-4E	10	4	40.0%
4	CCAFS LC-40	26	7	26.9%

Query 6: payload mass (kg) by site

```
SELECT LaunchSite, MIN(PayloadMass), AVG(PayloadMass), MAX(PayloadMass)
FROM LAUNCHES GROUP BY LaunchSite
```

LaunchSite	MinKg	AvgKg	MaxKg
VAFB SLC-4E	475.0	5814.0	9600.0
KSC LC-39A	2490.0	4377.0	6761.0
CCAFS SLC-40	362.0	3517.0	6092.0
CCAFS LC-40	0.0	2591.0	5271.0

Query 7: average payload by outcome

Outcome	Launches	AvgPayloadKg
Failure	32	3476.0
Success	24	3991.0

- KSC LC-39A ranks first (76.9%, 10 of 13); CCAFS LC-40 ranks last (26.9%, 7 of 26)
- VAFB SLC-4E carried the heaviest payloads (average 5,814 kg, maximum 9,600 kg); CCAFS LC-40 the lightest (average 2,591 kg)
- Successful landings carried a slightly higher average payload (3,991 kg) than failures (3,476 kg)

GitHub: https://github.com/RAHULKR123456/IBM-SpaceX-Capstone/blob/main/05b_EDA_SQL_Additional_Queries.ipynb

EDA with SQL Results (3): Payload Bands and Time Analysis

Query 8: success rate by payload band

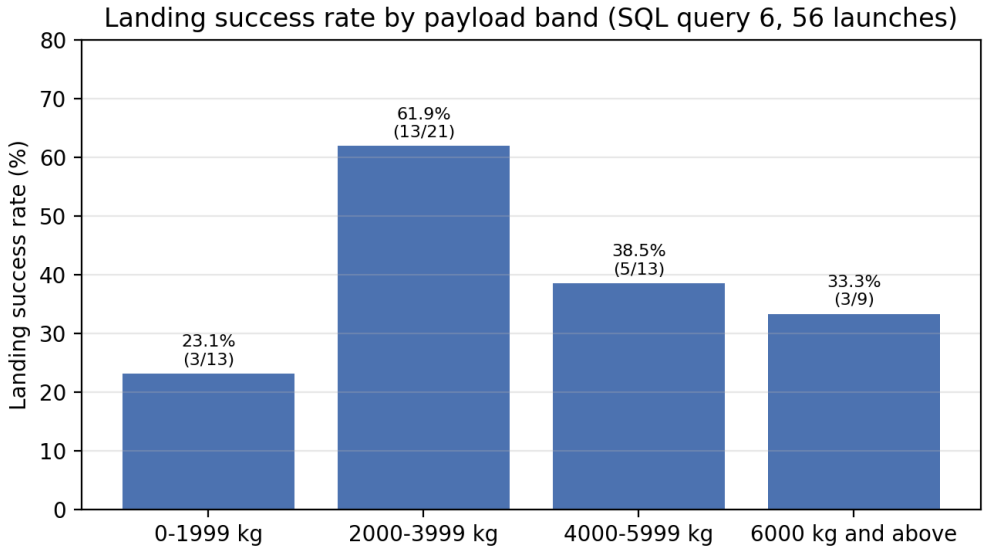
```
SELECT CASE WHEN PayloadMass < 2000 THEN '0-1999 kg' ... END AS PayloadBand,
COUNT(*), SUM(Class) FROM LAUNCHES GROUP BY PayloadBand
```

PayloadBand	Launches	Successes	SuccessRatePct
0-1999 kg	13	3	23.1
2000-3999 kg	21	13	61.9
4000-5999 kg	13	5	38.5
6000 kg and above	9	3	33.3

Query 9: time analysis

```
SELECT strftime('%Y', Date) AS Year, COUNT(*), ROUND(AVG(Class),3) FROM
LAUNCHES90 GROUP BY Year ORDER BY Year
```

Year	2010	2012	2013	2014	2015	2016	2017	2018	2019	2020
Rate	0.000	0.000	0.000	0.333	0.333	0.625	0.833	0.611	0.900	0.842

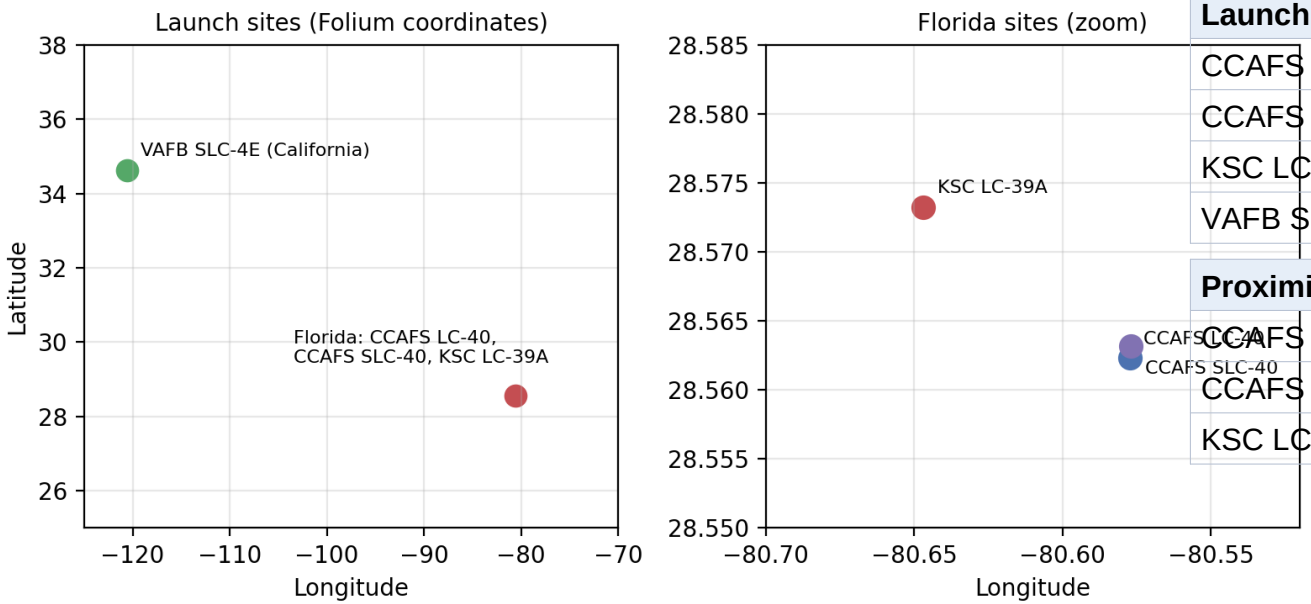


Success rate by payload band (output of Query 8)

- Success peaks for 2,000 to 3,999 kg (61.9%) and is lowest for the lightest launches (23.1%)
- Success does not rise steadily with payload mass
- Time analysis: the yearly trend is the strongest pattern

GitHub: https://github.com/RAHULKR123456/IBM-SpaceX-Capstone/blob/main/05b_EDA_SQL_Additional_Queries.ipynb

Folium Map: Launch Sites, Launch Records and Proximity



Launch site	Latitude	Longitude	Records	Success
CCAFS LC-40	28.56230	-80.57736	26	7
CCAFS SLC-40	28.56320	-80.57682	7	3
KSC LC-39A	28.57325	-80.64690	13	10
VAFB SLC-4E	34.63283	-120.61075	10	4

Proximity: site pair	Distance (km)
CCAFS LC-40 and CCAFS SLC-40	0.11
CCAFS LC-40 and KSC LC-39A	6.90
KSC LC-39A and VAFB SLC-4E	3,819.06

Launch-site locations from the Folium coordinates (interactive map: spacex_launch_map.html)

- 56 launch records shown as markers in a cluster: **24 green (success)** and **32 red (failure)** at four launch sites
- The three Florida sites lie within 7 km of each other; VAFB SLC-4E is about 3,820 km away on the California coast
- CCAFS LC-40 and SLC-40 are only 0.11 km apart: the same launch complex under two names

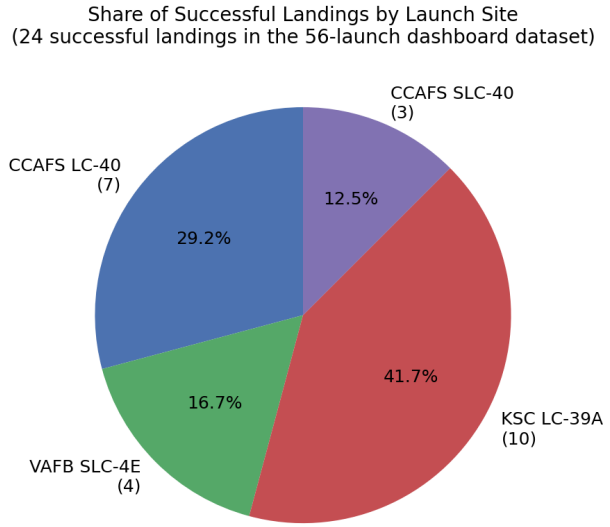
GitHub: https://github.com/RAHULKR123456/IBM-SpaceX-Capstone/blob/main/06_Folium_Interactive_Map.ipynb

Plotly Dash: Success Pie Charts

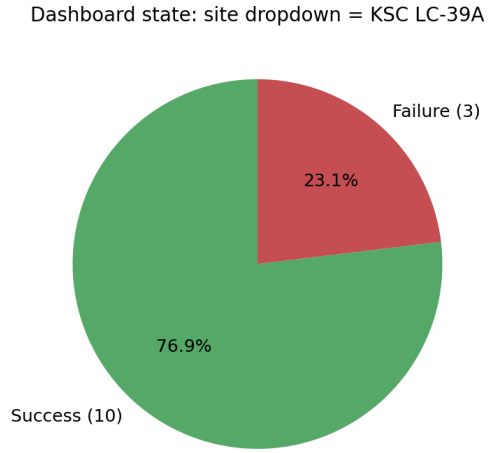
Dashboard components

Component	Behaviour
Site dropdown	All sites or one launch site
Pie chart	All sites: successes by site. One site: success vs failure
Payload slider	0 to 10,000 kg range filter
Scatter plot	Payload vs outcome, coloured by booster version

Launches: 56. Successful landings: 24 (CCAFS LC-40: 7, CCAFS SLC-40: 3, KSC LC-39A: 10, VAFB SLC-4E: 4).



Site dropdown = All Sites

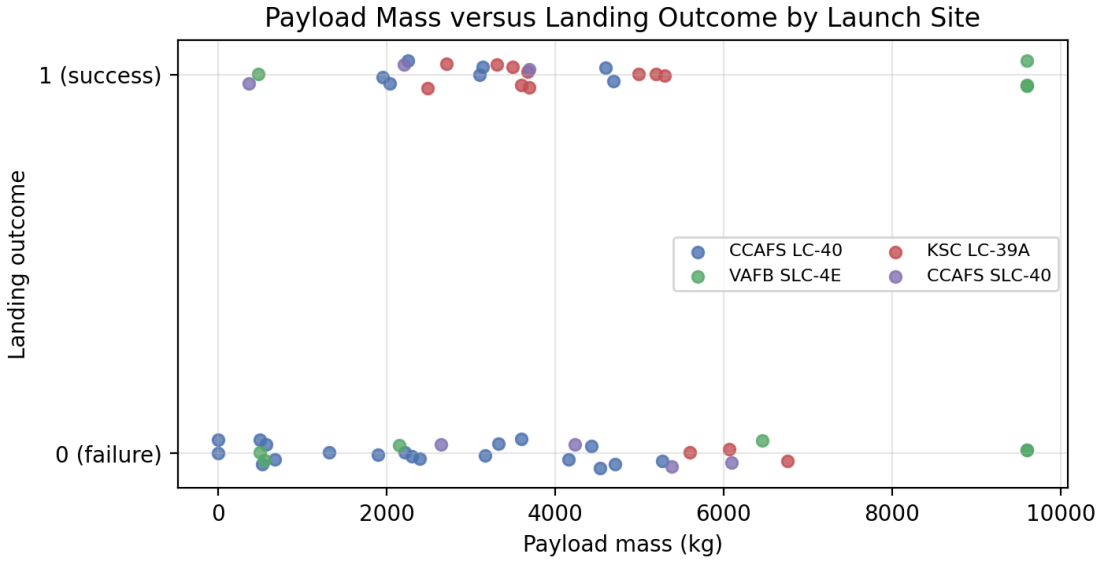


Site dropdown = KSC LC-39A

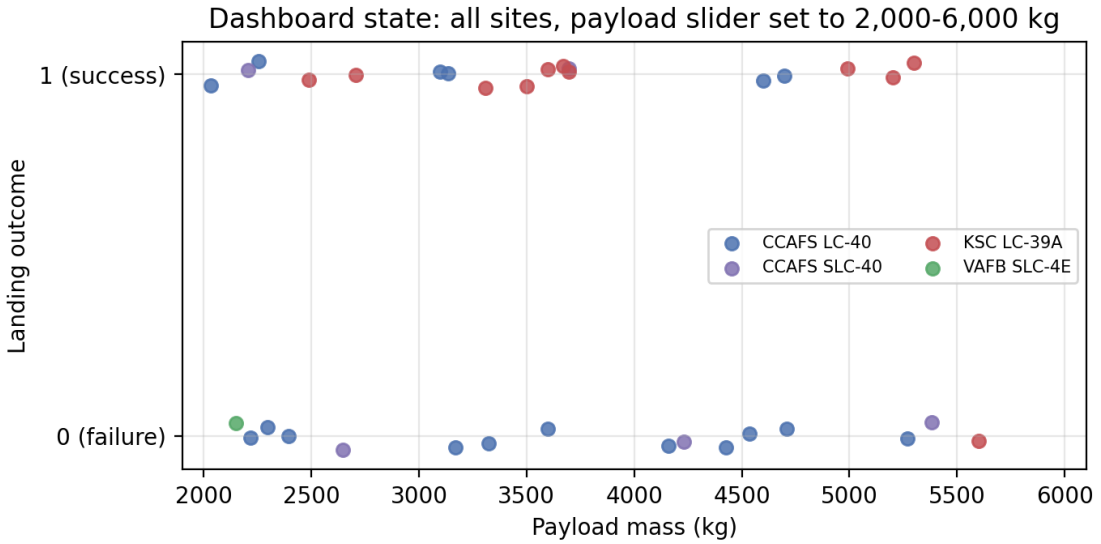
KSC LC-39A contributes the most successful landings (10 of 24, 41.7%) and has the highest within-site success rate (76.9%).

GitHub: https://github.com/RAHULKR123456/IBM-SpaceX-Capstone/blob/main/08_Interactive_Dashboard_Plots.ipynb | https://github.com/RAHULKR123456/IBM-SpaceX-Capstone/blob/main/app.py

Plotly Dash: Payload vs Launch Outcome



All sites, payload slider 0 to 10,000 kg

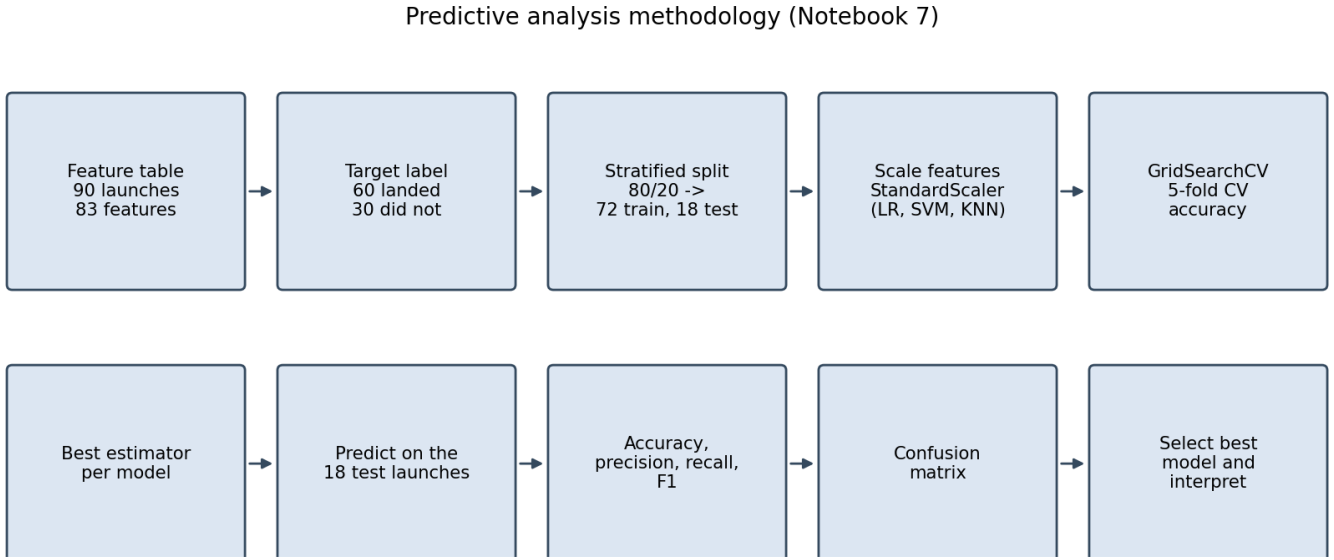


All sites, slider 2,000 to 6,000 kg: 34 launches, 18 successes (52.9%)

- Successful landings occur across payloads from about 360 kg to 9,600 kg
- All five launches carrying 9,600 kg were flown from VAFB SLC-4E; three of them landed
- Restricting the slider to 2,000 to 6,000 kg leaves 34 launches with a 52.9% success rate

GitHub: https://github.com/RAHULKR123456/IBM-SpaceX-Capstone/blob/main/08_Interactive_Dashboard_Plots.ipynb | https://github.com/RAHULKR123456/IBM-SpaceX-Capstone/blob/main/app.py

Predictive Analysis (Classification): Methodology



Predictive analysis workflow (Notebook 7)

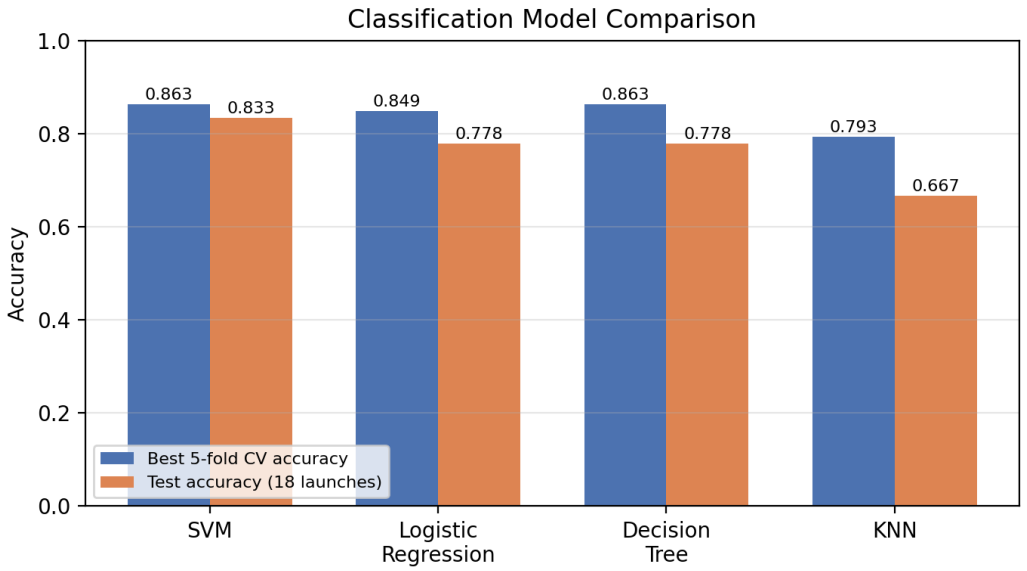
- 90 launches, 83 features; target: 60 landed, 30 did not
- Stratified 80/20 split: **72 train, 18 test**
- StandardScaler for Logistic Regression, SVM and KNN
- GridSearchCV, 5-fold CV, accuracy score
- Evaluation: accuracy, precision, recall, F1, confusion matrix

Model	Parameter grid	Best parameters
Logistic Regression	C: 0.01, 0.1, 1, 10	C = 0.1
SVM	kernel: linear, rbf, sigmoid; C: 0.1, 1, 10	sigmoid, C = 1
Decision Tree	max_depth: 2, 4, 6, 8, None; min_samples_split: 2, 5, 10	max_depth = 2, min_samples_split = 2
KNN	n_neighbors: 3, 5, 7, 9	n_neighbors = 5

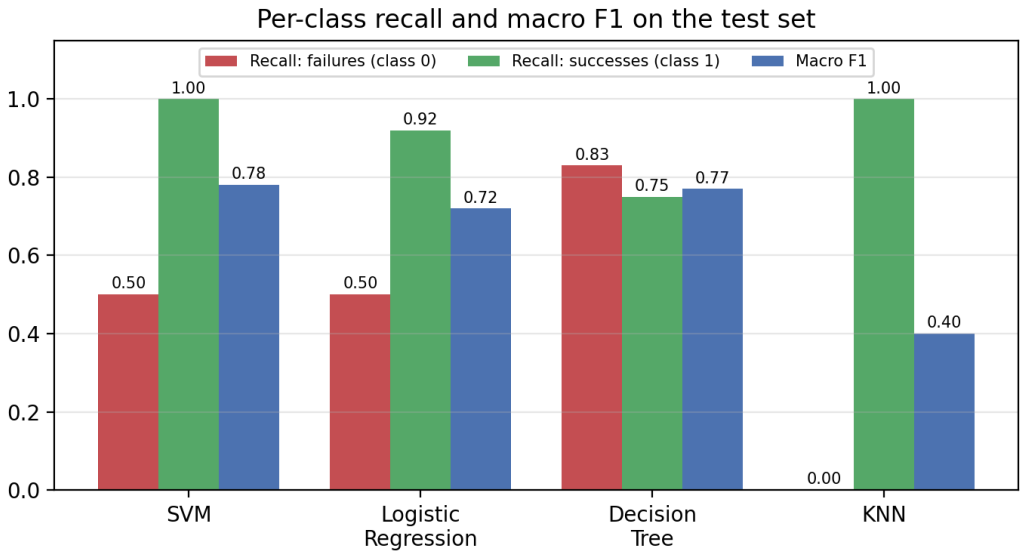
GitHub: https://github.com/RAHULKR123456/IBM-SpaceX-Capstone/blob/main/07_Predictive_Analysis.ipynb

Predictive Analysis Results: Model Evaluation

Model	CV accuracy	Test accuracy	Class 0 (failure) P / R / F1	Class 1 (success) P / R / F1	Macro F1
SVM (sigmoid, C = 1)	0.863	**0.833**	1.00 / 0.50 / 0.67	0.80 / 1.00 / 0.89	**0.78**
Logistic Regression	0.849	0.778	0.75 / 0.50 / 0.60	0.79 / 0.92 / 0.85	0.72
Decision Tree	0.863	0.778	0.62 / 0.83 / 0.71	0.90 / 0.75 / 0.82	0.77
KNN	0.793	0.667	0.00 / 0.00 / 0.00	0.67 / 1.00 / 0.80	0.40



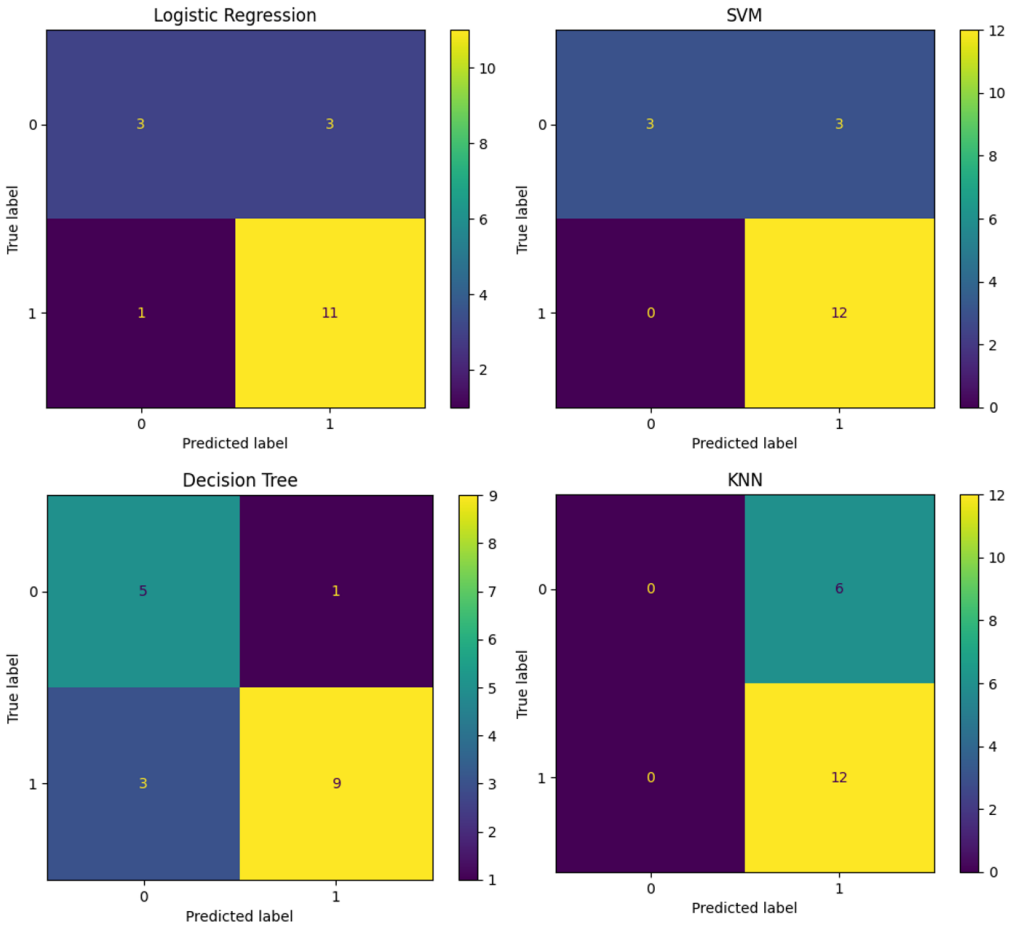
Cross-validation and test accuracy



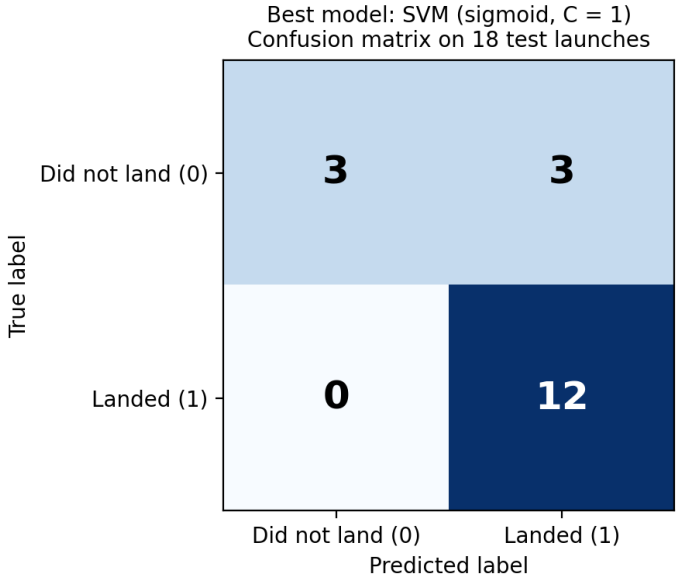
Per-class recall and macro F1

Test set: 18 launches (12 landed, 6 did not). Always predicting success scores 66.7%.

Predictive Analysis Results: Confusion Matrices and Best Model



Confusion matrices of the four models (18 test launches)

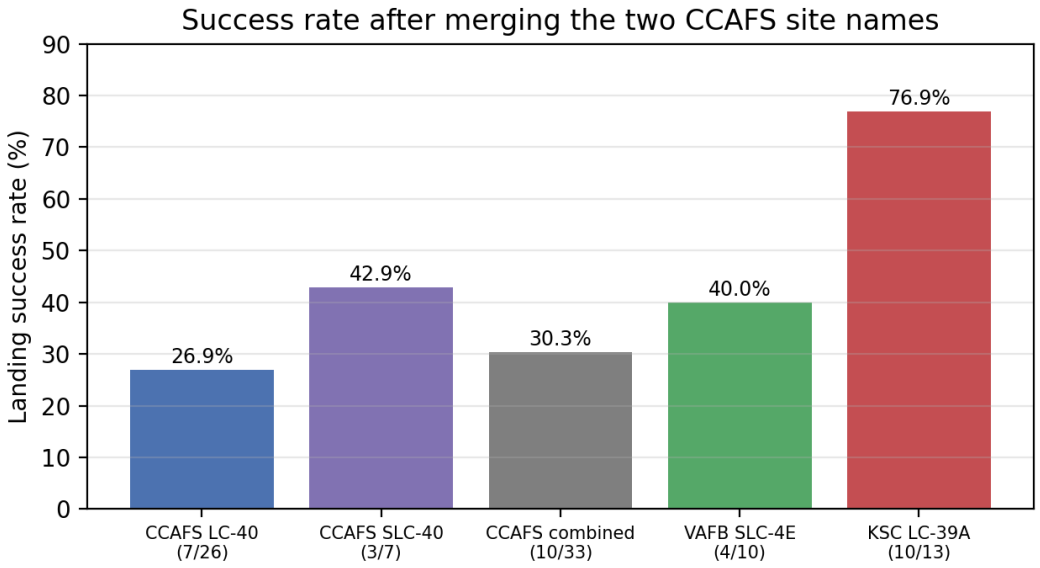


Best model: SVM

Best model: SVM

- **83.3% accuracy** (15 of 18 correct), highest macro F1 (0.78)
- Found **all 12** successful landings (recall 1.00)
- Flagged 3 of 6 failures (recall 0.50); its 3 errors are failures predicted as successes
- Decision tree finds 5 of 6 failures but misses 3 successes
- KNN predicts success for every launch (no failure detection)

Innovative Insights



Success rate after merging the two CCAFS site names (56 launches)

- **Two site names, one complex:** CCAFS LC-40 and SLC-40 are 0.11 km apart. Merged: 10 of 33 successes (30.3%), far below KSC LC-39A (76.9%)
- **Payload is not a simple dial:** success peaks in the middle band (61.9%) and the heaviest payloads came from one site
- **Accuracy hides the error type:** KNN scores 66.7% by always predicting success; macro F1 (0.40) exposes it
- **Time is the strongest signal:** much of the site difference reflects when each site was used

Conclusion

Summary of findings

- Landing success rose from 0% (2010 to 2013) to about 84% to 90% (2019 to 2020)
- KSC LC-39A ranks first (76.9%), CCAFS LC-40 last (26.9%), largely reflecting launch timing
- Payload mass alone does not separate outcomes; success peaks for 2,000 to 3,999 kg
- Three Florida sites lie within 7 km; VAFB SLC-4E is about 3,820 km away
- SVM is the best classifier: 83.3% test accuracy vs a 66.7% baseline

Recommendations

- Use time and site together when estimating landing risk
- Report per-class recall and the confusion matrix with any prediction
- Merge CCAFS LC-40 and SLC-40 before comparing sites

Future work

- Re-run data collection when the API is available
- Measure distance to coastline, railway, highway and city
- Repeated cross-validation, class weights and more features
- Host the Dash application online

Limitations: small datasets (90 launches; 18 in the test set) and a single train/test split, so the model ranking is indicative, not conclusive.

Project Repository and References

GitHub repository: https://github.com/RAHULKR123456/IBM-SpaceX-Capstone

Nine Jupyter notebooks and one Python application

File	Purpose
01_Data_Collection_API.ipynb	SpaceX API attempt
02_Data_Collection_Wikipedia.ipynb	Web scraping
03_Data_Wrangling.ipynb	Data wrangling
04_EDA_Visualization.ipynb	EDA charts
05_EDA_SQL.ipynb	SQL queries
05b_EDA_SQL_Additional_Queries.ipynb	Additional SQL
06_Folium_Interactive_Map.ipynb	Folium map
07_Predictive_Analysis.ipynb	Classification models
08_Interactive_Dashboard_Plots.ipynb	Plotly charts
app.py	Dash application

References

- IBM Skills Network, SpaceX capstone course datasets
- Wikipedia, List of Falcon 9 and Falcon Heavy launches (oldid 1027686922)
- SpaceX API v4 (request returned HTTP 525)
- Pedregosa et al. (2011), scikit-learn, JMLR 12, 2825 to 2830

Thank you